

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

A linear function f gives a company's profit, in dollars, for selling x items. The company's profit is \$220 when it sells 8 items, and its profit is \$320 when it sells 10 items. Which equation defines f ?

Answer

- A. $f(x) = 150x - 320$
- B. $f(x) = 32x$
- C. $f(x) = 50x - 10x$
- D. $f(x) = 50x - 180$

Correct Answer: D

Rationale

Choice D is correct. It's given that the relationship between x and $f(x)$ is linear. A linear function can be written in the form $f(x) = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that the company's profit is \$220 when it sells 8 items and the profit is \$320 when it sells 10 items. Since $f(x)$ represents the company's profit, in dollars, for selling x items, the graph of $y = f(x)$ in the xy-plane passes through the points (8, 220) and (10, 320). Substituting (8, 220) and (10, 320) for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{320 - 220}{10 - 8}$, which gives $m = \frac{100}{2}$, or $m = 50$. Substituting 50 for m , 8 for x , and 220 for $f(x)$ in $f(x) = mx + b$ yields $220 = (50)(8) + b$, or $220 = 400 + b$. Subtracting 400 from each side of this equation yields $-180 = b$. Substituting 50 for m and -180 for b in $f(x) = mx + b$ yields $f(x) = 50x - 180$. Therefore, the equation that defines f is $f(x) = 50x - 180$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.